Owen Westfold

owenwestfold@gmail.com · owenwestfold.github.io · linkedin.com/in/owen-westfold
Melbourne, Australia · Australian citizen

Mathematician and applied researcher. My background is in algebraic geometry, singularity theory and geometric representation theory. Alongside my work at RethinkX, I have been studying singular learning theory and its applications to neural networks, and am moving into AI alignment.

EDUCATION

PhD, Mathematics — University of Melbourne 2022
Supervised by Daniel Murfet.

MSc, Mathematics (with Distinction) — University of Melbourne 2018

BSc, Mathematics and Computer Science — University of Melbourne 2015

BA, Languages and History — University of Melbourne 2009

EXPERIENCE

Research Fellow — RethinkX 2023–present

RethinkX is an independent research organisation studying technology-driven disruption; much of my work here has been AI-related. Relevant projects have included:

- Technical advice on how AI works and on the trajectory of AI capabilities.
- Agent-based modelling of the macroeconomy, assessing the effect of AI and robotics on the labour market.
- Probabilistic forecasting of technology costs using Bayesian methods (ARIMA models in Stan).

Mathematics Tutor — University of Melbourne 2016–2022

RESEARCH

Doctoral. My thesis concerned the finite-dimensional irreducible representations of subregular finite W-algebras in simply-laced type. These algebras are quantisations of the Slodowy slice, whose intersection with the nilpotent orbit is a simple surface singularity resolved by the Springer map. Using Ginzburg's characteristic cycle construction, I showed the finite-dimensional irreducible representations are all one-dimensional. The work grew out of the Landau–Ginzburg / Conformal Field Theory correspondence.

Current. I am attempting to recast the asymptotic analysis underlying singular learning theory (Watanabe's grey book, Chapter 4) in the language of D-modules. So far, I have derived the asymptotic expansion of the state-density function directly from the V-filtration of the graph embedding.

PUBLICATIONS & NOTES

- O. Colman, A. Ghitza, N. Ryan. *Analytic evaluation of Hecke eigenvalues for Siegel modular forms of degree two.* ANTS XIII, Proceedings of the 13th Algorithmic Number Theory Symposium, Open Book Series **2**, No. 1 (2019), 207–220. (Published under my former surname, Colman.)
- *The asymptotic expansion of the state-density function via the V-filtration.* Independent note (draft; available on owenwestfold.github.io).
- PhD thesis, University of Melbourne, 2022 — on subregular finite W-algebras in simply-laced type. Minerva Access.
- Expository notes: derived categories and derived functors; sheaves on locally compact spaces; Hodge theory via the heat kernel.

TALKS

- May 2022 — *Quantising the nilpotent cone.* LG/CFT Seminar, Melbourne.
- February 2022 — *Finite W-algebras and the McKay correspondence.* LG/CFT Seminar, Melbourne.
- April 2021 — *The Springer resolution in type A.* LG/CFT Seminar, Melbourne.
- April 2021 — *Kleinian singularities and Slodowy slices.* LG/CFT Seminar, Melbourne.
- March 2021 — *Introduction to affine varieties.* LG/CFT Seminar, Melbourne.
- October 2020 — *What is Hodge theory?* What is...? Seminar (online).
- December 2019 — *DG Lie algebras and deformation theory.* Topology Student Seminar, Melbourne.

CONFERENCES & AWARDS

- The Mathematics of Conformal Field Theory II — ANU, Canberra (2021).
- Dean's Honours List, BSc — University of Melbourne (2014).

TECHNICAL SKILLS

Python, R, C · PyTorch, Keras · Stan, scikit-learn · pandas, tidyverse · AI-assisted workflows (Claude Code)